Predictive Models

- Machine Learning fundamentally differentiates itself from classical statistics by not assuming that the data comes from a specific model
- Hence ML, justifiably, can try different models on a given dataset to eventually pick the **best** one.
- Obviously, to do this one needs to define what it means to be **best.**
- Different "model performance measures" exist and any of these can be used to compare models - largely depending on the context and the kind of output:
  - Regression outputs are continuous numbers
  - Classification outputs are either
    - Class output (from algorithms like SVM and KNN that usually give a classification) or
    - Probability output (from algorithms like Logistic Regression, Random Forest that can give probability outputs)

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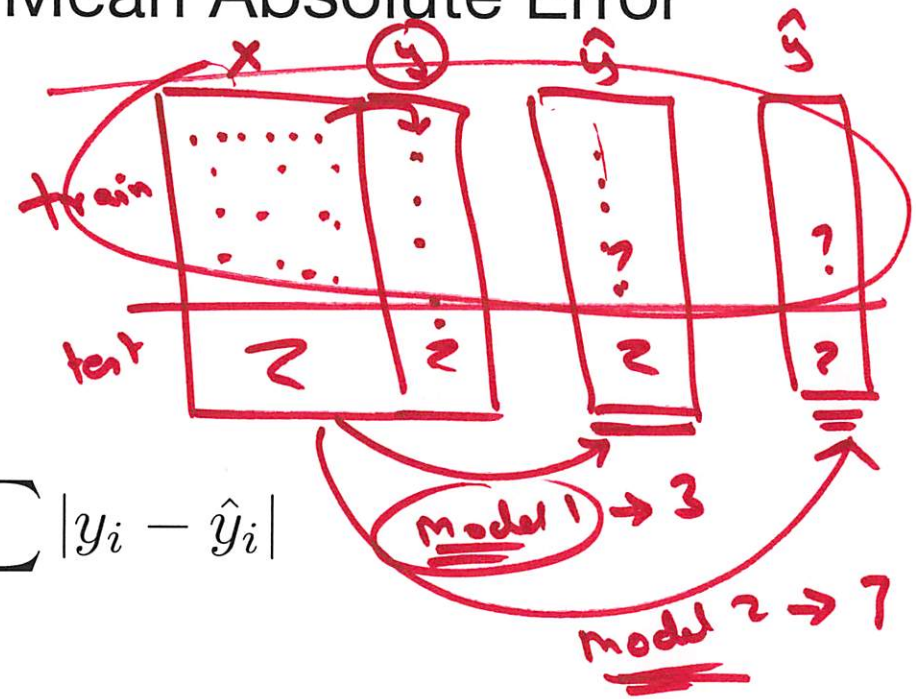
Root Mean Square Error, Mean Absolute Error

- For Regression Problems
  - Mean Absolute Error

$$\frac{1}{N}\sum|y_i - \hat{y}_i|$$

$$MAE = \frac{1}{N}\sum|y_i - \hat{y}_i|$$



  - RMSE (more justifiable when errors are assumed to be normal).

$$\sqrt{\frac{1}{N}\sum(y_i - \hat{y}_i)^2}$$

$$RMSE = \sqrt{\frac{1}{N}\sum(y_i - \hat{y}_i)^2}$$

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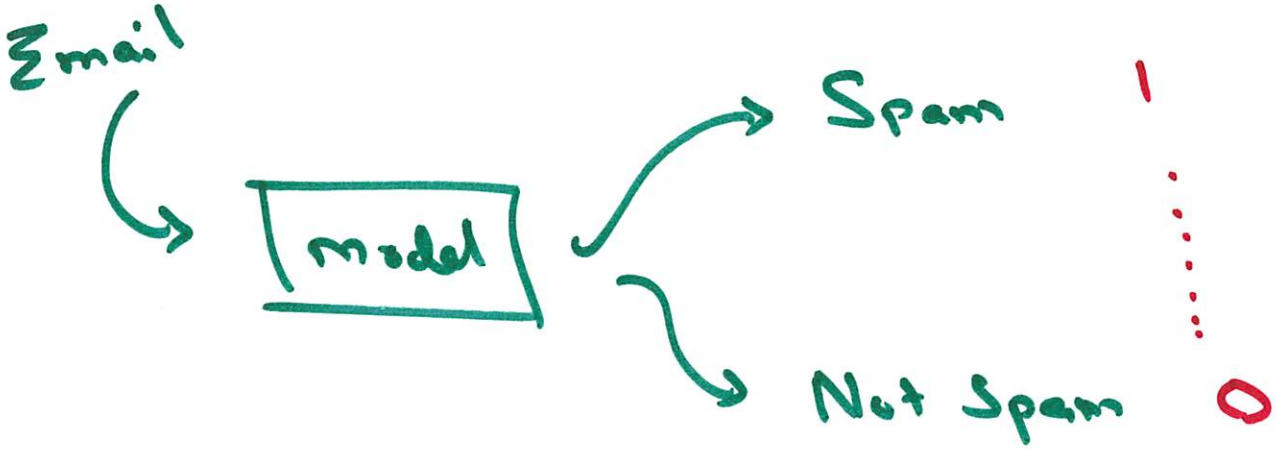

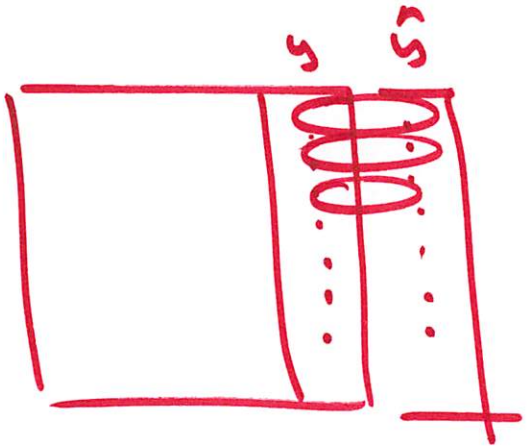

$$\cancel{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

loss =

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MAPE

$$\frac{1}{N}\sum\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

~~Model~~

m1 → 15 %

m2 → 35 %

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Cross Entropy Loss

$$\text{Loss} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

- For binary classification

$$\text{Loss} = -\left[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})\right]$$

when $y=1$

Prediction Type	Predicted $\hat{y}$	Cross-Entropy Loss	RMSE Loss
Confident & Correct	0.98	$-\log(0.98) \approx 0.020$	$\sqrt{(1-0.98)^2} = 0.02$
Confident & Wrong	0.02	$-\log(0.02) \approx 3.912$	$\sqrt{(1-0.02)^2} = 0.98$

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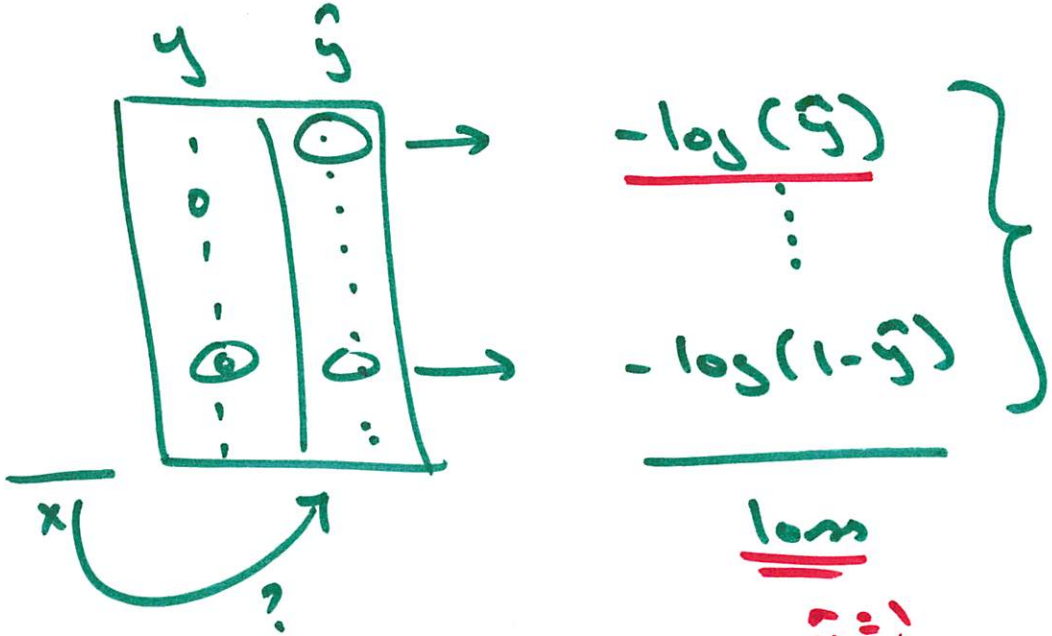

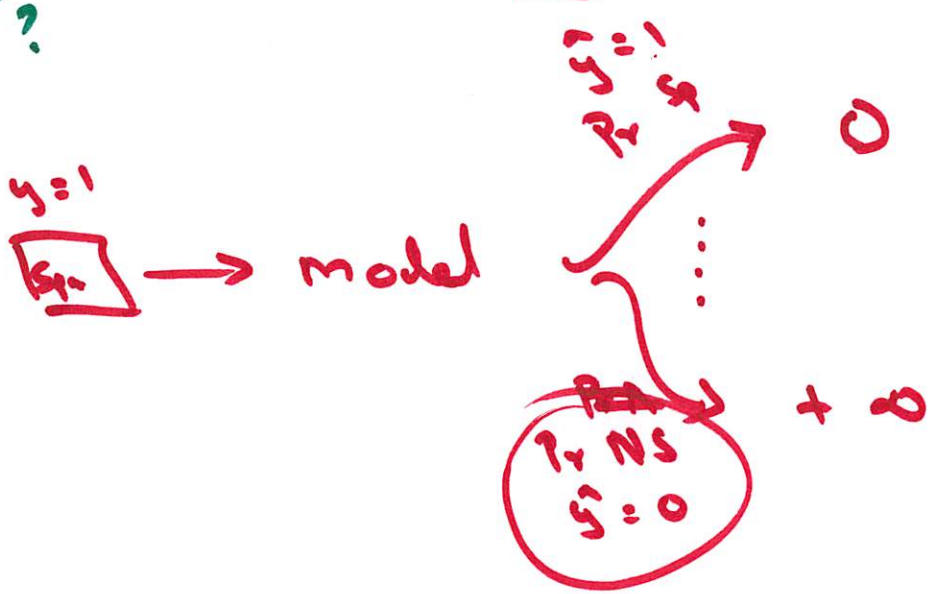

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Confusion Matrix

- For classification problem with a class output, the confusion matrix gives the counts of correct and erroneous predictions:

		Actual	
		1	0
Predicted	1	TP	FP
	0	FN	TN

- Classification Error Rate: sum of Type 1 (FP) and Type 2 (FN) Errors (in percentage). Accuracy is 1-(error rate)
- Sensitivity (also called Recall or True Positive Rate): Out of all the actual positive cases, how many did the model correctly identify.
- Precision: Out of all the positive predictions the model made, how many were actually correct.
- Specificity (also called True Negative Rate): Out of all the actual negative cases, how many were correctly predicted.

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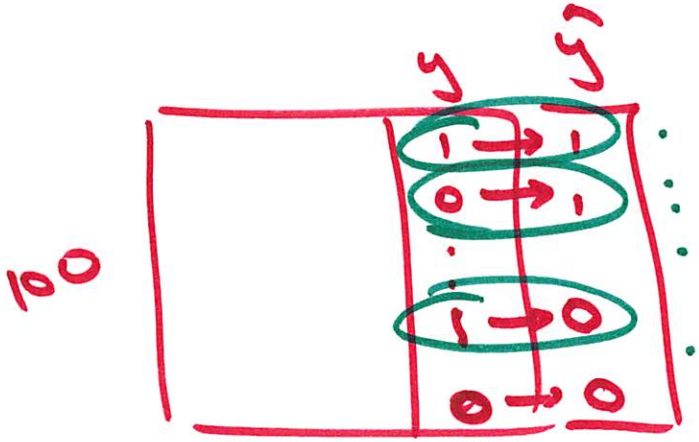

Predicted \ Actual	1	0
1	TP ✓ 40	FP × 3
0	FN × 7	TN ✓ 50

~~10~~ ~~990~~

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

~~Specificity~~

$$\text{Precision} = \frac{TP}{TP + FP}$$

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$$\text{Specificity} = \frac{TN}{TN + FP}$$

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$$F1 = \frac{2 \times P \times R}{P + R}$$

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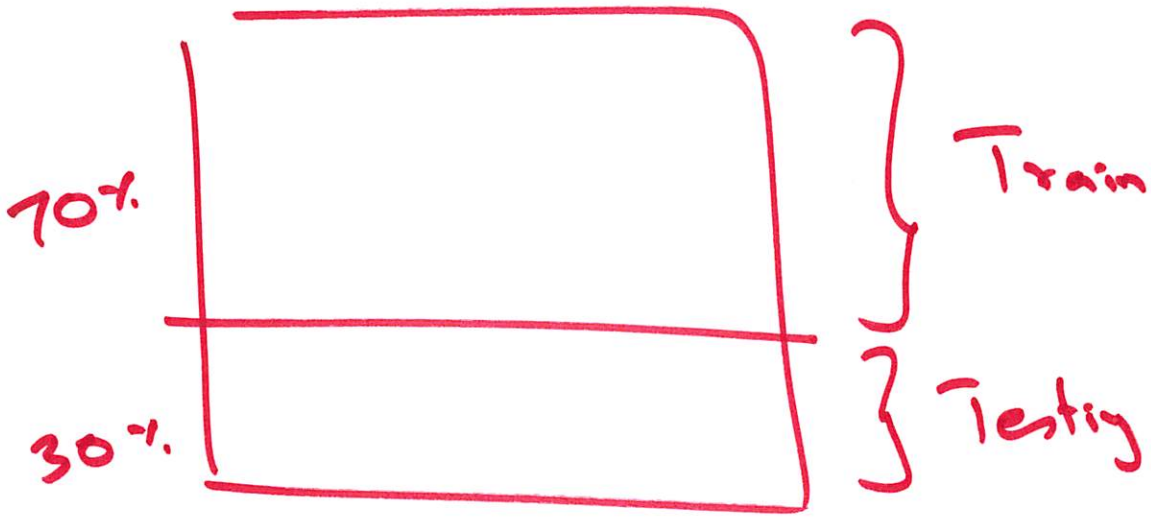

~~For~~ Cross Validation

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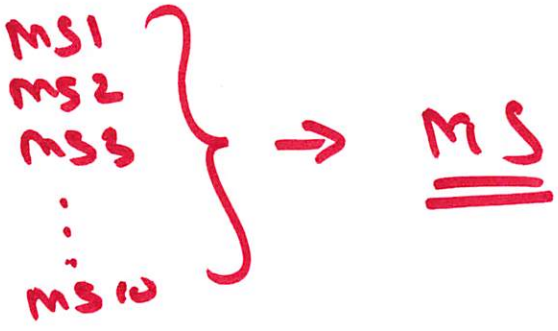
Performance on Testing Data

- Important to keep in mind that all these performance measures will improve on training data as the model becomes more complex
- So these performance measures are best compared on testing data.
- Cross Validation is a commonly used to reduce over-fit and improve performance on test data



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